

Introduction

Exercises ISLR – Ch.8

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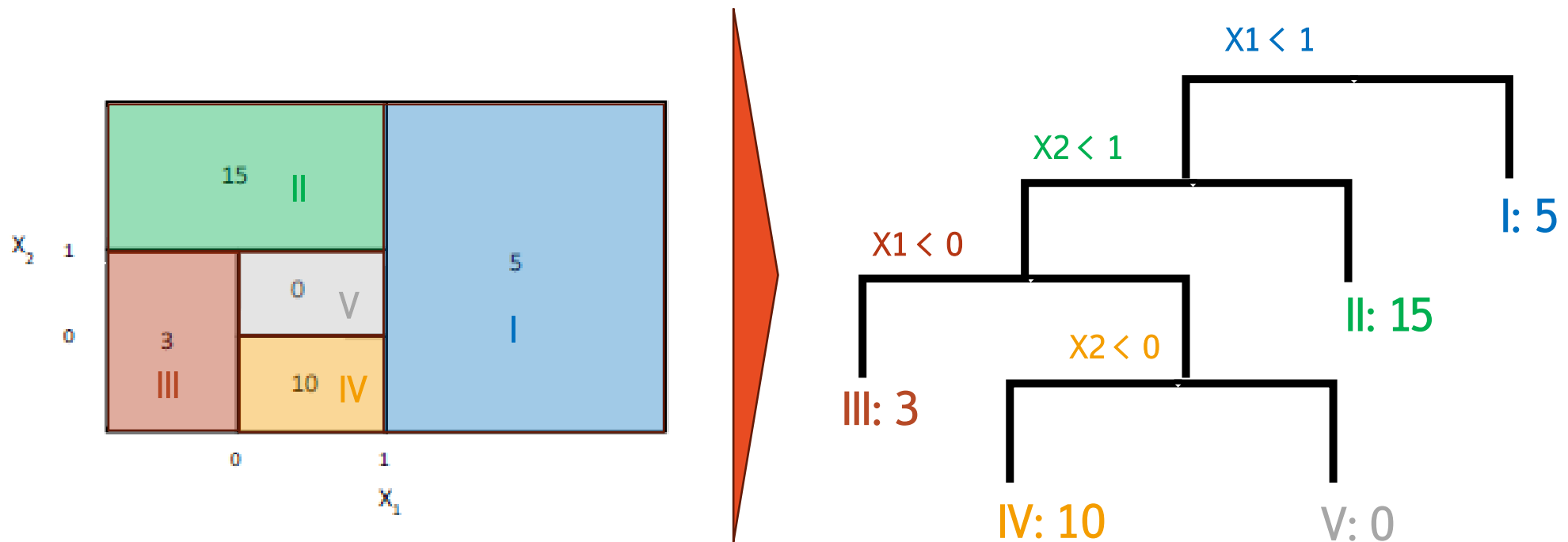


Conceptual

Exercise: 4

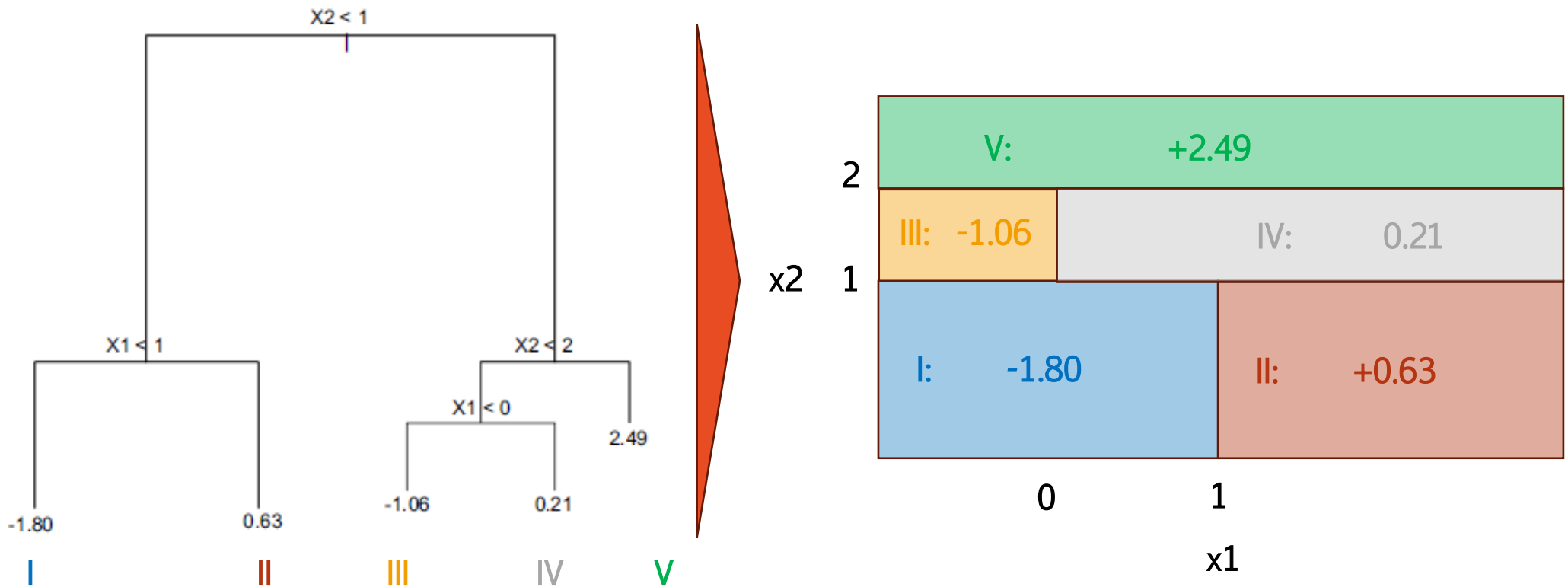
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Ex.4) Binary Tree – Space Partition



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Ex.4) Binary Tree – Space Partition





Applied

Exercise 13 – Boston dataset (regression)

Recap: Ch.6 - Crime Rate Regression



Full comparison based on MSE test data

Model <chr>	Test_MSE <dbl>
OLS: 13 predictors	48.97129
Lasso: 10 predictors	49.45321
Ridge: 13 predictors	49.96527
Backward: 4 predictors	50.44479
Forward: 2 predictors	51.84761
PCR: 5 components	53.95436

- **OLS with all predictors:** best performance even in test dataset, which means that the model is not suffering of overfitting despite using all predictors
- **Models with fewer predictors:** some of them presented a very good performance and could be selected to reduce the risk of overfitting in a different test data
- **Benefits of more parameters:** In this case, it seems that no variable or component emerge as being able to explain the behavior of crime rate

Recap: Ch.6 - Crime Rate Regression



Full predictors: 13 predictors

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	16.879384	8.215223	2.055	0.04068	*
zn	0.039526	0.021603	1.830	0.06819	.
indus	-0.066092	0.102269	-0.646	0.51855	
chas	-0.359512	1.359197	-0.265	0.79155	
nox	-11.391514	6.008571	-1.896	0.05882	.
rm	0.140098	0.691876	0.202	0.83965	
age	0.004565	0.020183	0.226	0.82121	
dis	-0.925751	0.317961	-2.912	0.00383	**
rad	0.539167	0.105247	5.123	5.05e-07	***
tax	-0.001345	0.006330	-0.213	0.83182	
ptratio	-0.251299	0.219012	-1.147	0.25202	
black	-0.007466	0.004083	-1.828	0.06837	.
lstat	0.138404	0.086771	1.595	0.11163	
medv	-0.148884	0.066666	-2.233	0.02618	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Residual standard error: 6.19 on 340 degrees of freedom

Multiple R-squared: 0.4709, Adjusted R-squared: 0.4506

F-statistic: 23.27 on 13 and 340 DF, p-value: < 2.2e-16

MSE – Test Dataset

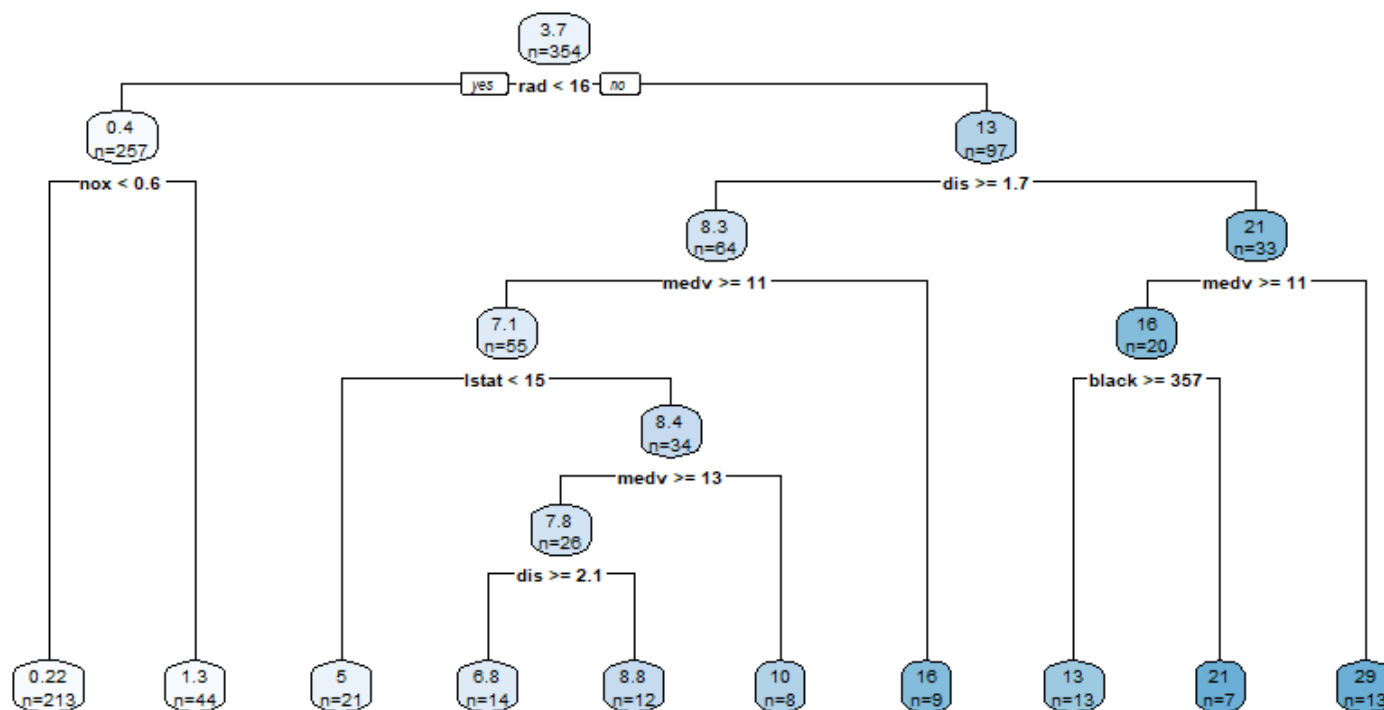
[1] 48.97129

Dis (-): distance to commercial centers
Rad(+): access to radial highways
Medv(-): median residence value

Ex.13) Regression Trees – Not Pruned



Decision Tree - Full (Not Pruned)



MSE – Test Dataset

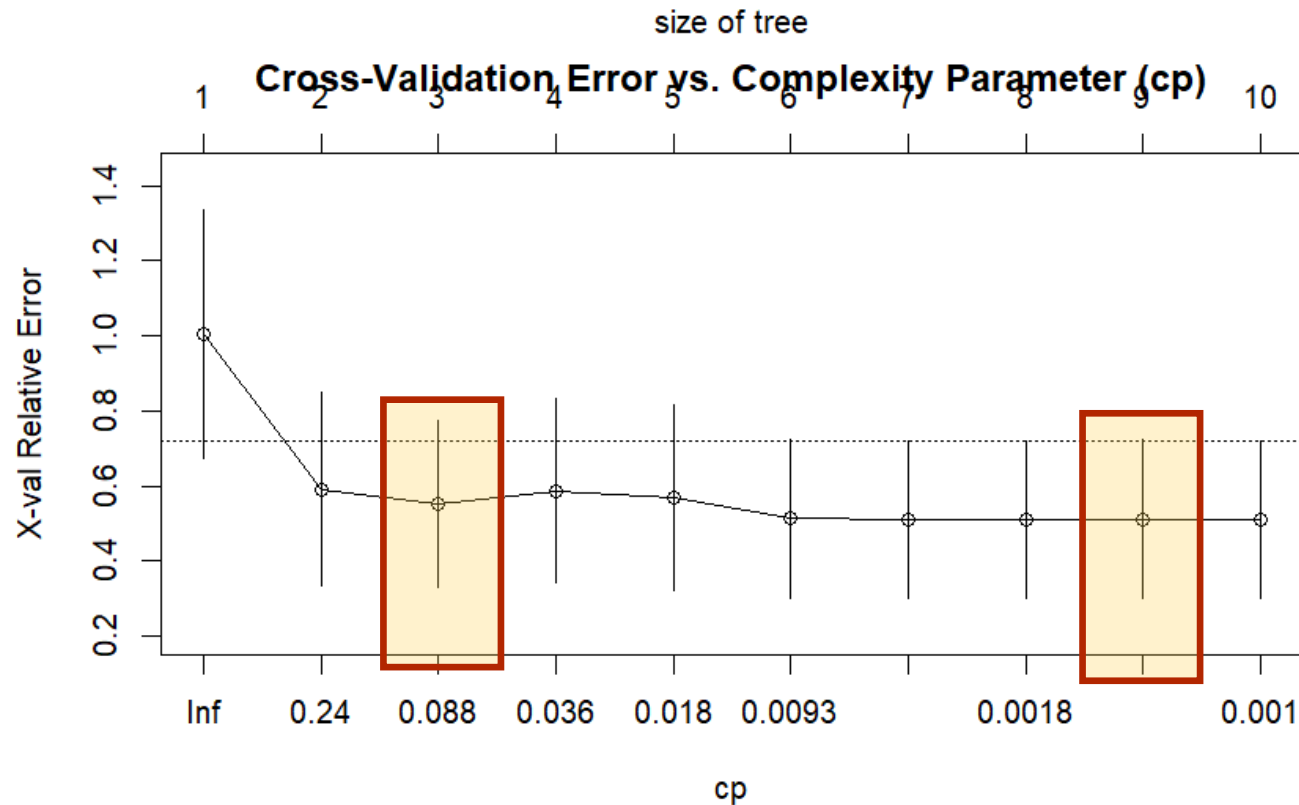
[1] 39.72773

Only 6 predictors out of 13 used to partition the space

- Rad
- Dis
- Medv
- Lstat
- Black
- Nox

10 Leaves – 9 divisions

Ex.13) Regression Trees – Not Pruned

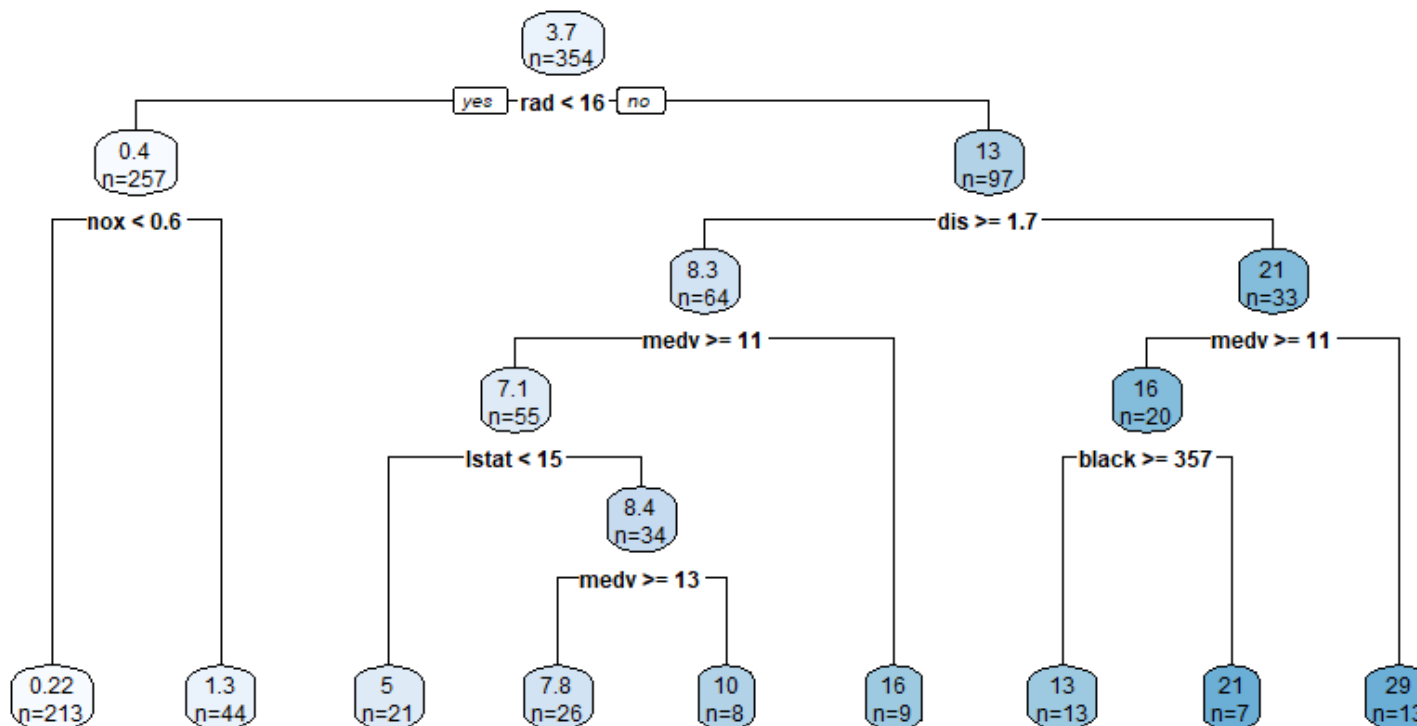


	CP	nsplit	rel error
1	0.4205639	0	1.00000
2	0.1389308	1	0.57944
3	0.0558725	2	0.44051
4	0.0227138	3	0.38463
5	0.0140850	4	0.36192
6	0.0061876	5	0.34783
7	0.0017817	6	0.34165
8	0.0017457	7	0.33986
9	0.0010294	8	0.33812
10	0.0010000	9	0.33709

Ex.13) Regression Trees – Best Pruned



Decision Tree - Pruned (Best CP)



MSE – Test Dataset

[1] 39.65202

Only 6 predictors out of 13 used to partition the space

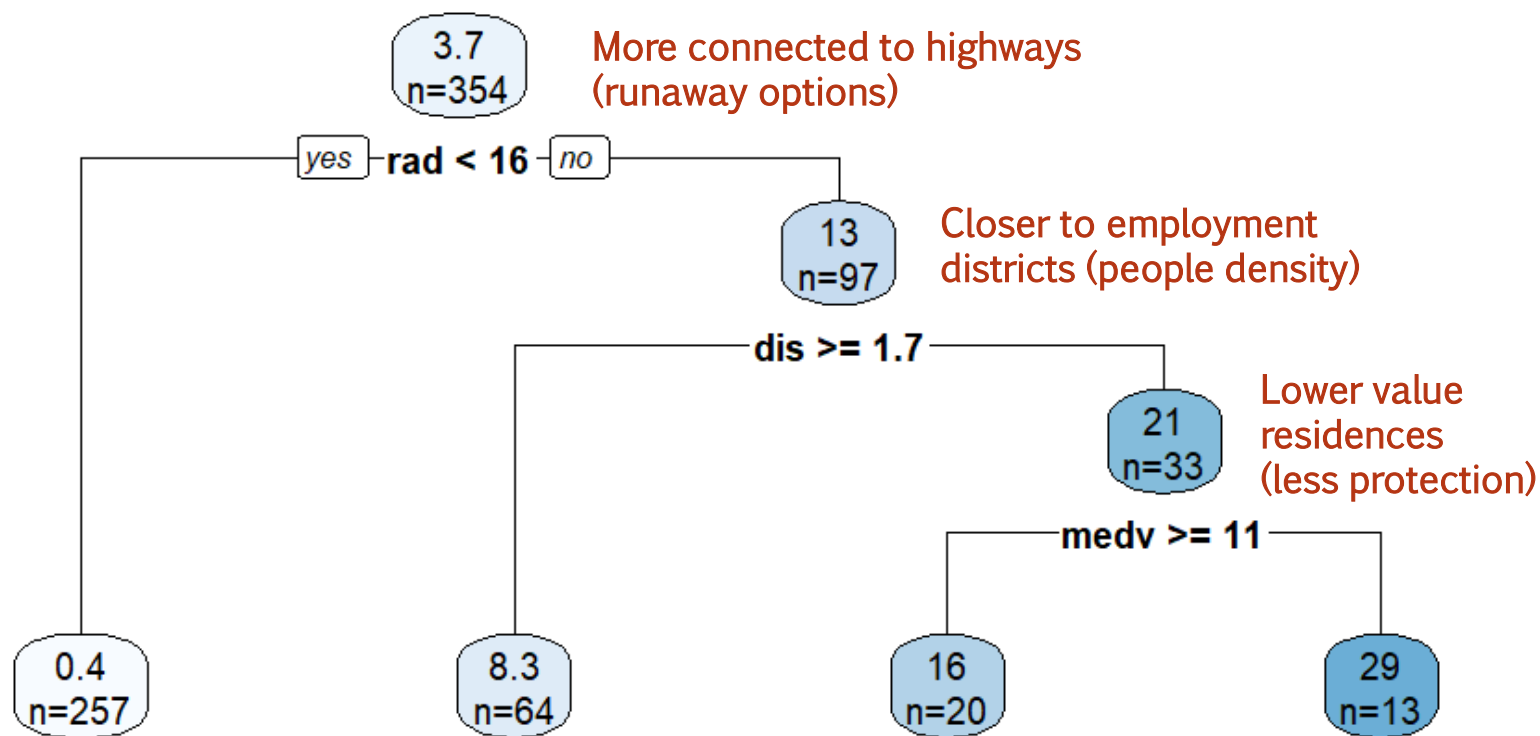
- Rad
- Dis
- Medv
- Lstat
- Black
- Nox

9 Leaves – 8 divisions

Ex.13) Regression Trees – Simple Pruned



Decision Tree - Pruned (3 Splits)



MSE – Test Dataset

[1] 45.69721

Only 3 predictors out of 13
used to partition the space

- Rad
- Dis
- Medv

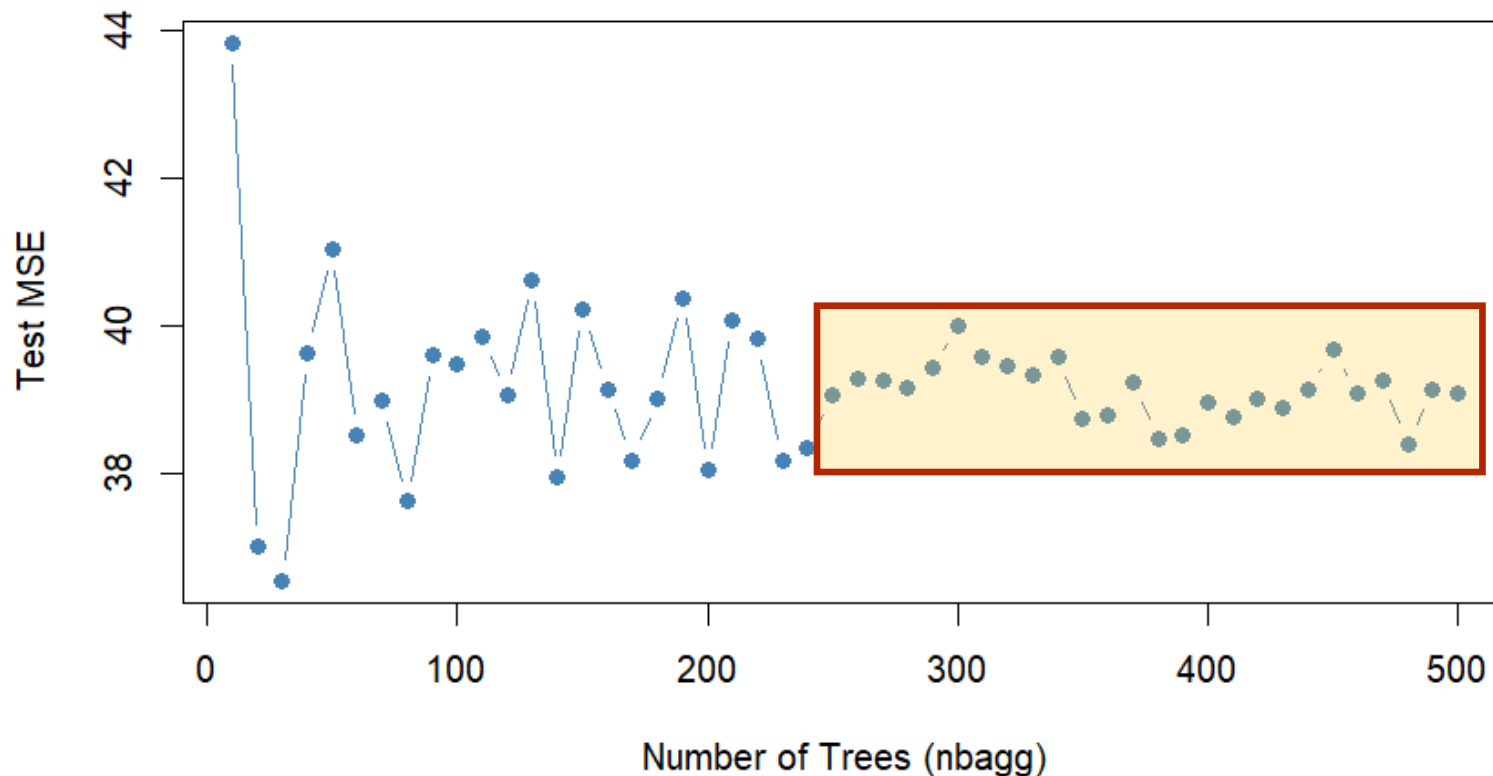
4 Leaves – 3 divisions

Improved interpretability

Ex.13) Regression Trees – Bagging



MSE vs. Number of Trees (Bagging)



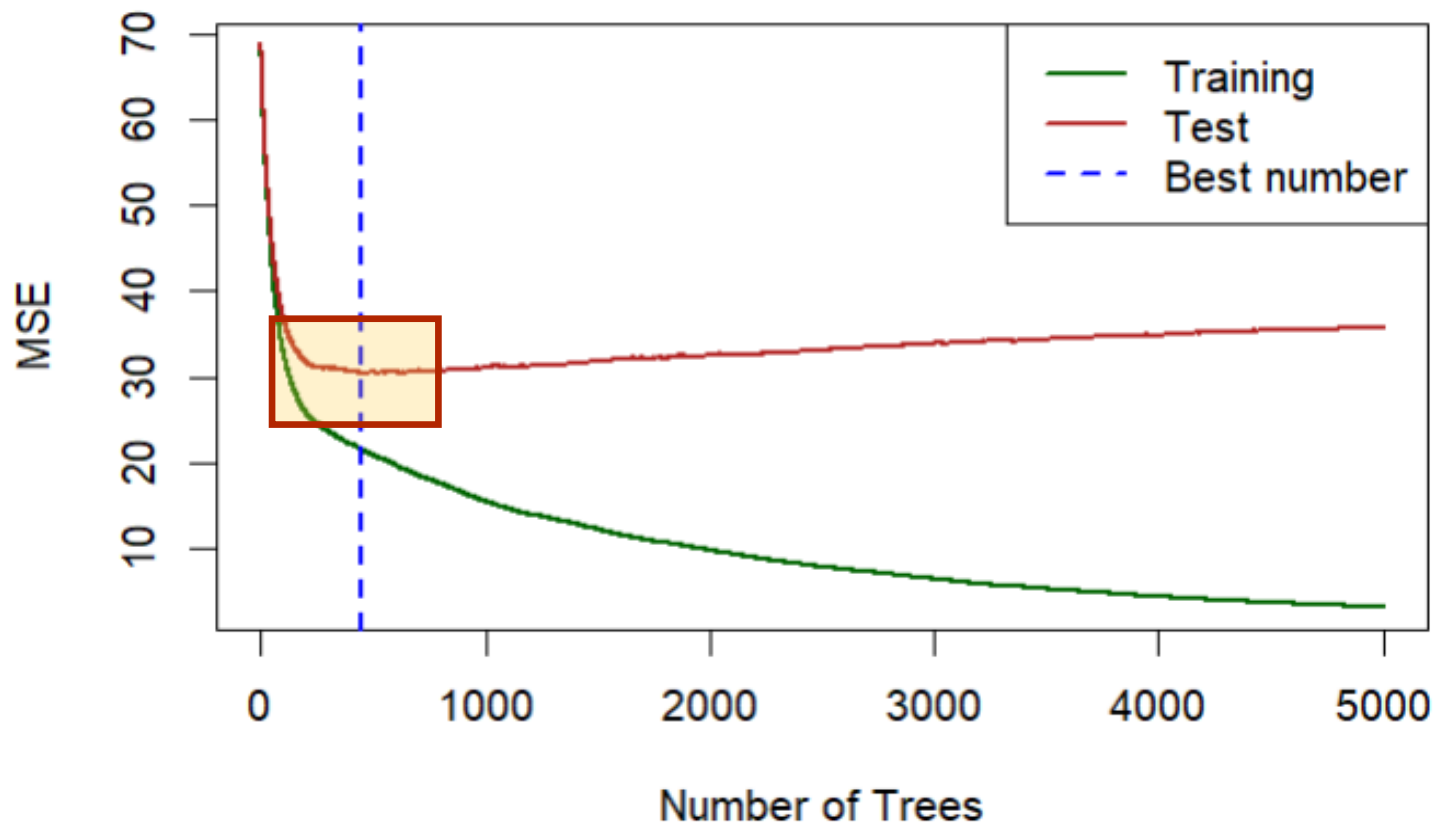
MSE – Test Dataset

[1] 39.14945

Ex.13) Regression Trees – Boosting



Error evolution (Boosting)

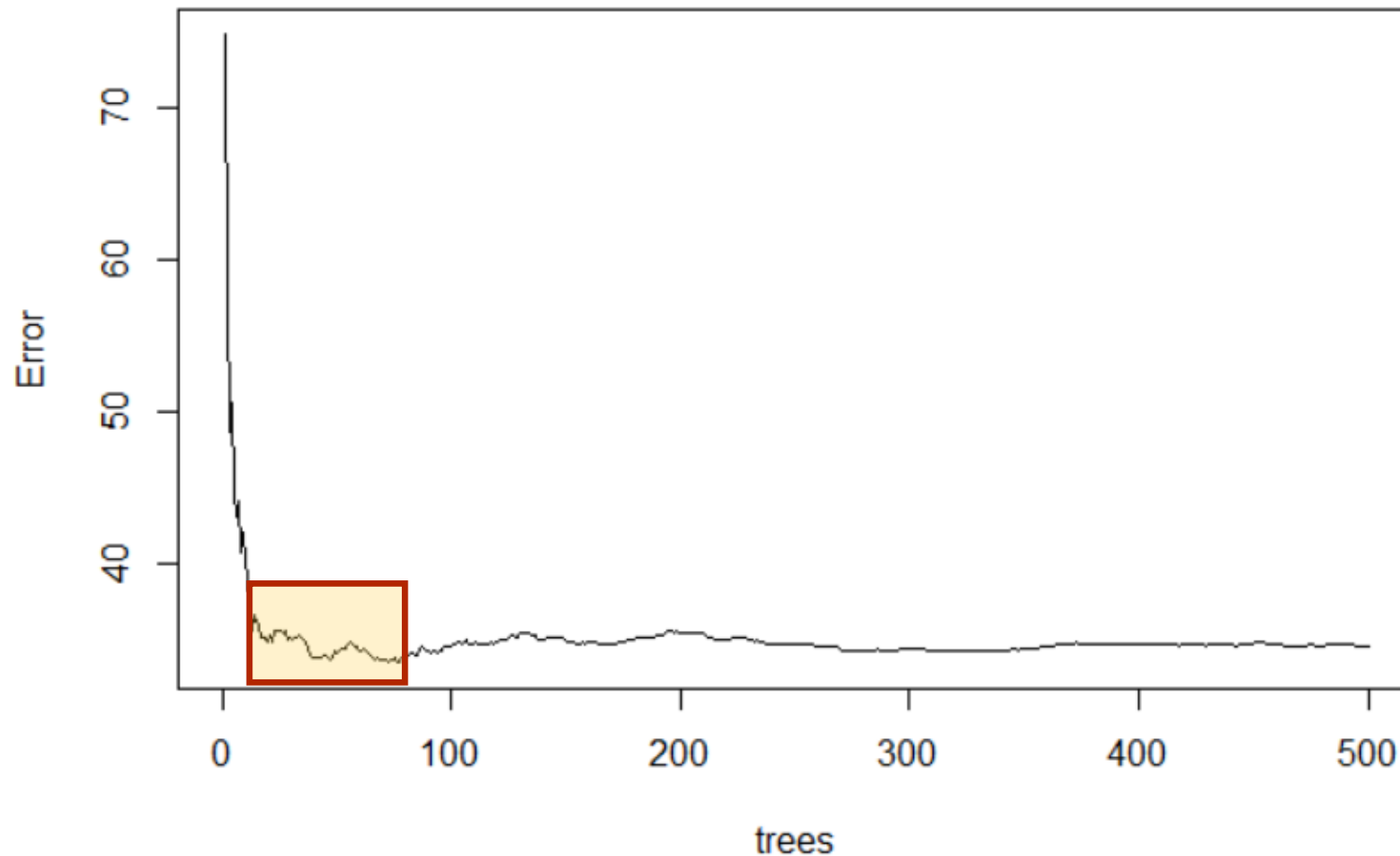


MSE – Test Dataset

[1] 37.44587

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Ex.13) Regression Trees – RandomForest



MSE – Test Dataset

[1] 36.88188

Ex.13) Regression Trees – Comparison



Model <chr>	Test_MSE <dbl>
RandomForest	36.88188
Boosting	37.44587
Bagging	39.14945
Decision Tree (Full)	39.72773
Decision Tree (Best Pruned)	39.80436
Decision Tree (3 Splits)	45.69721
OLS: 13 predictors	48.97129
Lasso: 10 predictors	49.50094
Ridge: 13 predictors	49.96527
Backward: 4 predictors	50.44479
Forward: 2 predictors	51.84761
PCR: 5 components	53.95436

- **Tree-based models outperformed traditional regression methods:** even a single decision tree surpassed the best linear models, effectively capturing non-linear relationships and interactions
- **Ensemble methods further improved accuracy:** combining multiple trees through bagging, random forests, or boosting significantly reduced variance and enhanced prediction, by leveraging the law of large numbers
- **Interpretability remains a key advantage:** although simpler trees underperformed compared to ensembles, they provided valuable insights into the main drivers of crime rate, offering a transparent view of variable influence